Advancing Seasonal Water Supply Forecasting for Lakes and Reservoirs Using Copulas

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Key Points:

- Vine copulas are introduced to seasonal lake water supply forecasting
- Representative model application to large lakes indicates high forecast skill
- Results accelerate advancement of seasonal-to-annual (S2A) forecasting

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Abstract

The future of water storage dynamics in lakes around the world is increasingly uncertain due to a changing climate and evolving anthropogenic demands. Now, more than ever, it is necessary to quantify the uncertainty in forecasts of future water supply and water levels. However, classical ensemble forecasting systems that rely on weighted repetitions of historical water flux sequences or independently resampled then post-processed sequences often fail to capture either the dependence between hydrological variables across space and time or the full range of plausible outcomes for these variables. In this paper, we outline an alternative statistical approach using high-dimensional distributions described by copulas to generate water supply ensembles which preserve the correlation structure among hydrological variables. We outline this approach in detail to support dispersed operational modeling efforts, demonstrated by one implementation of a copula-based model for seasonal to annual forecasting on the Laurentian Great Lakes. This model implementation performs similar or better than current operational forecasts while extending the forecast horizon from 6- to 12-months. Advantages and shortcomings of copula-based water supply modeling are discussed and future research directions are prescribed.

1 Introduction

Lakes are a vital resource for the planet. Among other beneficial uses, they host a range of natural habitats, serve as a foundation for human recreation and water supply, and are the medium for long- and short-range commercial marine transportation (Beeton, 2002). The extent to which lakes, and similarly functioning reservoirs, can support these services is modulated by the volume of water they store and, consequently, the areal extent and elevation of their surfaces and coastlines (Mortsch & Quinn, 1996; Hanson et al., 2002; Marchand et al., 1988). However, future changes in global lake storage and lake water level dynamics (including changes in lake shoreline extent and accessibility) are increasingly uncertain under dual pressures imposed by climate change and anthropogenic demand (Yao et al., 2023). Therefore, understanding historical and potential future changes in global lake (and reservoir) water supplies across multiple time and space scales is critical to human and ecosystem health and resilience. At the heart of this understanding and forecasting capability lies the development, testing, and application of climate and hydrological models focused specifically on the water balance of lakes and reservoirs.

Lake water balance models aim to simulate changes in lake storage (and water level) by quantifying all inflows and outflows (referred to as water balance components) including over-lake fluxes (i.e. via precipitation and evaporation), lateral tributary runoff, groundwater fluxes, and (in some cases) inflows and outflows through connecting channels (Winter, 1981). Like most land surface hydrological models, lake water balance models range in complexity, from process-based to empirical (or semi-empirical) (Ozdemir et al., 2023). For example,L. Sun et al. (2020) developed a coupled physics-based climate, land-surface, and 3-dimensional (3-D) hydrodynamic modeling system to forecast water levels, while Barzegar et al. (2021) and Coulibaly (2010) applied machine learning directly to a water level time series (actually circumventing the water balance approach) to accomplish the same goal.

Lake water balance forecasting systems based entirely on physics, however, are often biased, especially over long (i.e. seasonal to multi-decadal) time horizons due (in part) to the complexities of representing changes in lake thermal structure, surface temperature, ice cover, and evaporation (Bürger et al., 2011; Nahar et al., 2018). Physics-based forecasting systems are also, understandably, more computationally expensive and require a high level of process understanding to run effectively. Both limitations constrain their geographic distribution, the length of their forecasting horizons, and the ability to generate probabilistic or multi-member ensemble forecasts. Similarly, while purely statistical (e.g. machine learning) methods can be computationally efficient and simple to

implement, they also can require specific content expertise to be developed and implemented correctly. Furthermore, machine learning models are often somewhat inflexible to user adjustments (e.g. operator specification of a certain scenario) and face credibility challenges due to a lack of both interpretability and physical basis of model components, leading to frequent characterization of these models as "black boxes" that are not favored in operational settings (Weber et al., 2012). Although, it is worth noting that advances in human-in-the-loop processes (user supervised machine learning models) appear to provide a promising compromise (Zhou et al., 2021).

As an alternative to both highly process-based and purely statistical methods, many lake water balance forecasting systems are based on some variation of a conventional ensemble approach in which either historical meteorological forcings are propagated through simple hydrological models, or historical water balance component sequences are propagated into the future as potential realizations of future water supplies (Weber et al., 2012). Advantages of utilizing historical data as a basis for lake water supply forecasting include a robust reflection of historical variable dynamics (such as marginal probability distributions, seasonal patterns, and frequency of extremes) and interdependencies (including temporal and spatial cross- and auto-correlations). However, these approaches draw, by definition, on a limited range of historical sequences, determined by the length of the historical water balance record, that may not reflect the full range of potential hydrological outcomes under either a stationary or non-stationary climate (Troin et al., 2021). Resampling methods, including both parametric and non-parametric approaches (such as bootstrapping), have been employed to reflect variable distributions (Khatri et al., 2018; Robinson & Loáiciga, 2006; Fortin et al., 1997; H. Li et al., 2023), but require post-processing (reordering) of samples to capture interdependencies (Lall & Sharma, 1996; Clark et al., 2004). These methods increase in complexity with the dimension of dependence (Srivastav & Simonovic, 2015; King et al., 2014).

In this study, we address these contemporary challenges facing lake water balance modeling by introducing and testing a novel copula-based seasonal-to-annual (S2A) scale ensemble water supply forecasting system which draws upon the strength of statistical approaches while maintaining a physical basis and comprehensible structure. Copulas are a family of multivariate probability distributions that encode multivariate dependence structures and accommodate arbitrary marginal distributions of individual variables (Czado & Nagler, 2022). Copulas allow specification of any marginal probability distribution for each variable, rather than simply utilizing Gaussian marginals (a common approach in multivariate statistical analysis). Importantly, the parameters of the marginal distributions and the multivariate dependence structure can be modified to reflect potential changes in future climate conditions.

Copula-based modeling has gained popularity in water resources management applications after Bárdossy (2006) first used copulas to capture spatial dependence in groundwater quality parameters (Tootoonchi et al., 2022). In the decade and a half since, copulas have been used to study (among other hydrologic science topics) flood and drought frequency (Singh & Zhang, 2018; F. Li & Zheng, 2016; Borgomeo et al., 2015; Madadgar & Moradkhani, 2014), rainfall-runoff relationships (Abdollahi et al., 2019; Xiong et al., 2014), multi-site streamflow (L. Chen et al., 2015; Worland et al., 2019; Samaniego et al., 2010), agricultural yields (Yang et al., 2023; Ribeiro et al., 2019), and climate model downscaling (C. Sun et al., 2021; Verhoest et al., 2015). While a handful of recent studies (Huang et al., 2023; Gu et al., 2023; Perz et al., 2022) have applied copulas to aggregated water balance components, we are unaware of studies, other than those taken as a stepping stone to our current study in VanDeWeghe et al. (2022), that attempt to fully resolve correlation between multiple water balance components of a lake (or system of lakes) across space and time. Vine copulas are a uniquely well-suited tool for this very task due to their flexibility to accommodate high dimensional data while maintaining interpretability and capacity for user specification. In this paper, we present a detailed

explanation of how copulas can be applied specifically to water supply forecasting and demonstrate the utility of this approach with a case study.

2 Methods

In the following sections, we describe our approach to generalizing a vine-copula-based modeling framework for ensemble forecasting of lake and reservoir water balance components. We begin with background on the mathematics of copula functions, followed by customization of our framework in the development of a new seasonal-to-annual (S2A) water supply forecasting system for large lakes and reservoirs (including multi-lake and multi-reservoir systems with high spatial correlation). We then provide a representative application to the Laurentian Great Lakes for which we describe our approach to acquiring historical data for fitting the copula function, followed by our approach to testing and validating the new copula forecasting system.

2.1 Copulas: Methodological Overview

A copula is a joint distribution function that quantifies the relationship between marginal distributions of a suite of variables and a multivariate cumulative distribution function. A key feature of copulas is that any multivariate distribution can be expressed through a combination of a copula (the term ascribed specifically to the dependence structure among model variables), and the variables' $(x_1, ..., x_d)$ marginal distributions, which are transformed to be uniform on [0,1]. Mathematically, this relationship can be expressed as (Sklar, 1959):

$$F(x_1, ..., x_d) = C(F_{x_1}(x_1), ..., F_{x_d}(x_d))$$

There are many families of copulas, each designed to capture different dependency and tail structures between marginal distributions (Nelsen, 1997; Huard et al., 2006; Czado & Nagler, 2022; Frahm et al., 2003). However, some of the most simple copula formulations, such as the Archimedean or elliptical families (including Gaussian), are most applicable to bivariate or low-dimensional data and are hindered by tail dependency or symmetry requirements, inflexible parameterization, or intractability at high dimensions (Oh & Patton, 2017). Two alternative copula constructions are the vine and factor copulas (factor copulas are a special case of vine copulas, but often treated separately) which utilize bivariate distributions as building blocks for higher dimensional distributions (Bedford & Cooke, 2001, 2002; Krupskii & Joe, 2013). While vine copulas generally require more complex parameterization (scaling with the square of the dimension) than factor copulas (scaling linearly with the dimension), vine copulas avoid an explicit assumption of latent variable dependency in the factor method (Krupskii & Joe, 2013). Factor copulas have historically been applied to financial datasets (performance indices, stock returns) and spatiotemporal environmental datasets representing a single process across relatively small domains (e.g. regional wind speed, temperature, or humidity) that are more readily attributable to small set of latent variables than the interrelated but unique and complex processes of multiple water balance components across broad temporal (e.g. seasonal to annual) and spatial (e.g. continental-basin) scales (Krupskii & Joe, 2013; Krupskii et al., 2018; Krupskii & Joe, 2015; Krupskii & Genton, 2019; Oh & Patton, 2017). We advance with application of vine copulas for this study, though it is plausible that simpler copula forms, such as factor copulas, may be found in future research to be sufficient for capturing the relationships between water balance components for lake systems with a relatively small spatial domain.

The most flexible form of vine copula is the regular (R-) vine, which contains the simplified cases of the cannonical (C-) and drawable (D-) vines (Bedford & Cooke, 2001, 2002; Aas et al., 2009; Dißmann et al., 2013). The first level, or tree, of a vine is com-

posed of nodes which are distributions and edges which are bivariate copulas fit between two of the nodes. For the following tree, each edge from the previous tree becomes a node (using the copula to the describe a new distribution). This is repeated for $d-1$ trees until a single distribution conditioned on all of the marginals is described. C-vines have one node at each tree that has edge connections to all other nodes, and D-vines have nodes that are never connected to more than two other nodes. For a graphical representation of vines and more detailed review of vine methods, see Czado and Nagler (2022). Both of these constructions reduce the complexity of the vine, but imply specific dependence structures. Due to the high-dimensionality of time- and space-varying water balance components and lack of a known dependence structure (such as a single key variable or ordered dependence), we do not impose the constraints of C- or D- vines, although the constructed R-vine retains the flexibility to resemble both.

2.2 Using Vine Copulas

In the following four subsections, we address the fitting of the vine copula structure, the selection of parametric marginal distributions, random sampling from the copula distribution, and a sampling algorithm to select ensemble members 'matching' initial conditions.

2.2.1 Fitting Dependence Structures

Each variable $x_1, ...x_d$ should be approximately uniformly distributed on [0,1] to fit a copula (Nelsen, 1997). To meet these criteria and move from the native distribution and units of time-integrated water volume for each variable, we apply an empirical probability integral transform, where x_{ij} is the i-th observation ($i \in [1...n_{observed}]$) of the j-th variable ($j \in [1...d]$), u^*_{ij} is its standard uniform counterpart, and r_{ij} is the rank of x_{ij} among all observations of the same variable, ordered from least to greatest.

$$u^*_{ij} = r_{ij}/(n_{observed} + 1)$$

A copula can now be used to describe the joint distribution of these uniformly distributed marginals. Specification of this copula (in the case of vines) requires selection of (1) a vine structure (arrangement of pair-copulas), (2) pair-copula families, and (3) parameters for each pair copula (Dißmann et al., 2013). Following the algorithm of Dißmann et al. (2013), the vine structure is constructed by maximizing the dependence between variable pairs (measured by Kendall's tau) at each tree. Thus, representing the relationship between highly dependent variables (which would exert more influence on the fit of the distribution) is prioritized. The copula family is selected for each variable pair, at each tree, by identifying the copula form which maximizes the AIC (Akaike, 1973). Parameter values for each pair copula are selected by maximizing the log-likelihood over all parameters, following Aas et al. (2009).

2.2.2 Selecting Marginal Distributions

Many algorithms have been proposed that purport the most robust estimation of marginal distributions for copula modeling (Singh & Zhang, 2018; Silva & Lopes, 2008; X. Chen et al., 2006; Karmakar & Simonovic, 2008), any of which may be appropriate given specific model objectives and modeler experience with the statistical methods from which they draw. The general goal of these algorithms is to select the marginal distributions and parameters that lead to the best-fitting overall copula model. For clarity, the marginal distributions which we use in our case study are based in previous Great Lakes water balance models, and are discussed further in Section 2.3.3.

2.2.3 Sampling from Vine Copulas

To simulate plausible sequences of water supply, component values are sampled at points on the multivariate distribution described by the vine copula. Specifically, sampling is implemented using the inverse Rosenblatt transform (Rosenblatt, 1952), a process which we describe only briefly here, adapted from Aas et al. (2021). A random point $u' = (u'_1, ..., u'_d)$ is specified on the d-dimensional unit hypercube $[0,1]^d$ over which the distribution is described by a copula. For the j-th variable in $(j = 1, ..., j = d)$, u'_j is independent of all $u'_{\overline{j}}$ and is standard uniform. Then, values for the j-th variable on $[0,1]^d$ which reflect the dependency structure of the distribution F can be obtained from the inverse Rosenblatt transform where

$$u_1^{*\prime} = F^{-1}(u'_1);\ u_2^{*\prime} = F^{-1}(u'_2|u'_1);\ ...;\ u_d^{*\prime} = F^{-1}(u'_d|u'_1, ..., u'_{d-1})$$

and F^{-1} is performed through numerical integration. Then, $x_j^{'} = F_{x_j}^{-1}(u_j^{*\prime})$ where x'_j is a sample value of the j-th variable in units of water volume, $u_j^{*\prime}$ is the corresponding $[0,1]$ value on F, and $F_{x_j}^{-1}$ is the quantile function of the parametric marginal distribution of the j-th variable. This procedure is repeated to generate n randomly sampled water balance component sequences, $x_{ij} \in \mathbb{R}^{n \times d}$, which follow the distribution F, $x_{ij} \sim F$.

2.2.4 Approximating Initial Conditions

Sampling from random locations on the distribution yields an ensemble of plausible water balance component sequences which are representative (in distribution, correlation, and dynamics) of the complete calibration period. In the case of subseasonal to annual forecasting, known initial conditions of the system reduce the distributional space from which plausible samples should be drawn. However, conditional sampling from high-dimensional distributions described by R-vines is still an ongoing area of research in the statistical community and (to our knowledge) has only been implemented for copulas formed by constrained vine structures (typically a subset of C- or D-vine), rather than for an arbitrary R-vine (Aas et al., 2021; Aigner et al., 2023; Bevacqua et al., 2017; Jobst et al., 2023; Kraus & Czado, 2017). Future possibilities with conditional sampling are discussed further in Section 4. As a proxy for conditional sampling, we implement a selection algorithm that 'matches' initial conditions from randomly sampled sequences with observed initial conditions.

Given known values, $x_{tj \in (1,...,m)}$ for m variables at the time t of a forecast, a sample should be drawn which 'matches' $x_{tj \in (1,...,m)}$ and will contain forecast values for the remaining $d-m$ variables. A selection of n' 'matched' samples forms an ensemble forecast that mimics the result of sampling from a distribution of the forecast variables conditioned on the m known variables. Here, we perform this matching using just one of many possible algorithms by more heavily weighting samples which minimize the difference between sampled and known initial values relative to the inherent uncertainty in the estimation of each known value,

$$z_{ti} = \sum_{j=1}^{j=m} \frac{x'_{ij} - x_{tj}}{\overline{\sigma'_j}}$$

where z_{ti} is the matching score of the i-th random sample for a forecast at time t, x'_{ij} is the value of the j-th variable of the i-th sample, x_{tj} is the known value of the j-th variable at time t, and $\overline{\sigma'_j}$ is the average standard deviation of all probabilistic estimates of the j-th variable over the calibration time (if probabilistic estimates exist) or another reasonable estimate of the uncertainty in x_{tj}. Each sample is then weighted in a new en-

semble based on its relative matching score, where, for example, the sample with the highest (worst) z_{ti} is included once and the the sample with the lowest (best) z_{ti} is included $\frac{max(z_{ti})}{min(z_{ti})}$ times. Then, the 'matched' ensemble forecast is formed by drawing n' samples from the weighted set.

2.3 Representative Application

In this study, we implement construction and sampling of an R-vine copula following the methods (described above) of Aas et al. (2009) and Dißmann et al. (2013) using the `rvinecopulib` package (Nagler & Vatter, 2023) within the R statistical programming environment (R Core Team., 2023).

2.3.1 Study Area: Laurentian Great Lakes

The Great Lakes collectively represent one of the largest surface freshwater systems on the planet (Gronewold et al., 2013) and support a range of commercial (Millerd, 2011), industrial (Florida, 2020; Campbell et al., 2015), and recreational (Nevers & Whitman, 2011) activities for the 40 million inhabitants of its watershed (MacKay & Seglenieks, 2013). The Great Lakes system is connected by channels which route flow from Lake Superior, through Michigan-Huron, Erie, and Ontario to the St. Lawrence River. While these connecting channel flows represent major fluxes in the water balance of each lake, they are readily modeled by hydraulic stage-fall discharge relationships (being only a function of water levels) and thus are subject to much different uncertainty than the remaining water balance components, which we classify as water supply (Thompson et al., 2020).

Previous short-term (subseasonal to annual) water supply forecasting efforts on the Great Lakes have been limited in their capacity to 1) forecast at the annual time-scale, 2) represent initial conditions, 3) incorporate anticipated climatic conditions, and 4) quantify forecast uncertainty. Operational Great Lakes water supply or water level forecasting tools, described in Fry et al. (2020) and Gronewold et al. (2011), are not used to forecast beyond 6-months. Furthermore, no single operational product robustly addresses more than two of the challenges identified above. Several research oriented machine learning models have also been used to generate water level forecasts (Wang & Wang, 2020; Demir & Yaseen, 2023; Coulibaly, 2010). Generally, these models more explicitly recognize initial conditions (through time-lagged variable dependencies) but lack uncertainty quantification or climate flexibility. In this case study, we build upon VanDeWeghe et al. (2022), which demonstrates a climate-dynamic copula model, to establish the ability of copula-based water balance models to represent initial conditions and uncertainty in a seasonal to annual Great Lakes water supply forecast.

2.3.2 Fitting the Copula to Historical Data

For each discrete lake (Superior, Michigan-Huron, Erie, Ontario) we calculate the water supply that is independent of connecting channel flow as the summation of the three largest water balance components: over-lake precipitation, evaporation, and runoff (Gronewold et al., 2021). Historical monthly estimates of these component values are taken from results of the Large Lake Statistical Water Balance Model (Gronewold et al., 2016, 2020; Do et al., 2020), which uses a Bayesian framework to assimilate many estimates of each component from independent data products and subsequently infer probabilistic estimates of monthly component values, constrained to close the water balance from observed lake levels. For clarity, and to represent a more typical implementation of copula-based water balance modeling, we extract only the median values to use as historical component estimates. We use the longest suitable period of record available (1951-2021) to both calibrate (80%, 1951-2006) and validate (20%, 2007-2021) our copula model.

2.3.3 Great Lakes Copula Model

The input variables are discretized by component, lake, and month (relative to the forecast month). Due to both the hydraulic connectivity of the system (upstream water supply propagates downstream) and the likelihood of regional climate factors influencing multiple lakes, we select a model which jointly forecasts water balance components for each lake at each month. To ensure that each member of a 12-month ensemble forecast represents an internally consistent hydroclimate realization, the model must capture temporal, spatial, and intervariable correlation across the entire 12-month forecast horizon. Likewise, a unique copula model is calibrated for each month of the calendar year, so that a forecast produced in any month will retain correlation across the entirety of the next 12-month horizon. The extent to which lagged component values are correlated to values in the forecast month and beyond will determine how many additional months (prior to the 12-forecast months) will be included to represent initial conditions. Therefore, each copula model consists of 4(lakes)$\times$3(components)$\times$(12+k)(months) variables, where k is the number of months included prior to the forecast month.

Following the precedent established in historical Great Lakes water balance reconstructions described in Section 2.3.2, we specify normal, log-normal, and gamma marginal probability distributions for (respectively) monthly evaporation, runoff, and precipitation on all lakes. Fitting and sampling of each month's copula model was performed as specified in Sections 2.2 and 2.2.3. For this demonstration, n' and n were fixed at 1000 and 100,000, respectively for computational efficiency, but increasing n would improve how closely 'matched' samples follow the theoretical conditional distribution.

3 Results

An analysis of, and comparison between serial correlation in historical and copula-simulated Great Lakes water balance components (Figure 1) indicates that, generally, the copula reproduces the serial correlation of the historical data. We find further that it does so especially for water balance components, and at lag intervals, in which autocorrelation in the historical data is relatively high (Mackay et al., 2021; Fleming & Sauchyn, 2013). The strong (based on a threshold of 0.5) serial correlation in winter and summer runoff on Lake Superior and (to a slightly lesser extent) on Lake Ontario over lag periods ranging from 1 to 3 months, for example, is evident in the historical record and is reflected in our copula simulations, shown in Figure 1. This finding reflects the value of vine-structure fitting algorithms that prioritize variables with strong dependence. It is also informative to note that runoff across the Great Lakes in winter months is relatively low, and therefore correctly reflecting serial correlation in spring and summer runoff will likely have the greatest relative benefit to seasonal forecasting.

Given the importance of reflecting serial correlation in historical and future water supply information (Anghileri et al., 2016), our findings support the adoption of copula-based models into the suite of water resources management planning and forecasting frameworks and (for frameworks designed for water supply forecasting across annual horizons) the use of a 15-month copula calibration window. To be clear, this window includes a 3-month retrospective period used as a basis for 'matching' forecasts with observed initial basin conditions and a subsequent 12-month forecast period. It is informative to note that, in future applications of our copula-based forecasting framework, any antecedent window could be selected for a particular study area depending on serial correlation in water supplies, and the desired forecast horizon.

Results of our analysis also indicate that the copula reproduces spatial correlation (for our study, across each of the Great Lakes) between total water supply (Figure 2) and between individual water supply components (results not shown explicitly, but implicit to recreating water supply). In some cases, we find that the spatial correlation struc-

ture in historical water supply data is quite complex; the ability of copulas to capture complex spatial (and serial) correlation structures further underscores their potential utility in advancing S2A hydrological forecasting.

For example, we find that there is a strong bimodal correlation structure between cumulative water supplies on Lake Superior and Lake Erie (left panel, Figure 2) that reflects changes in cumulative water supply relationships across different seasons. This correlation structure also reflects the subtle but interesting bimodal marginal distribution of Lake Superior cumulative water supplies (we might expect, for example, that cumulative water supplies for a lake system would have a unimodel distribution). Near-average values of water supply on Lake Erie are correlated with both high (summer) and low (winter) values on Lake Superior. The correlation structure between Lake Erie and Lake Ontario cumulative water supplies (right panel, Figure 2) is also strong but, unlike the Lake Superior - Lake Erie correlation structure, is unimodal and non-linear.

Importantly, our results suggest that cumulative water supply projections across S2A scales that do not account for these spatial and temporal dependencies (particularly on the Great Lakes, but likely on other lakes as well) might systematically either over- or under-predict extreme water supply (and water level) events. Of course, directly propagating historical water supply sequences into the future (a conventional approach that, as noted earlier, is still routinely used in operational forecasting environments, including those on the Great Lakes) implicitly reflects system-wide spatial and temporal correlation, but is limited by the number of historical sequences and the complications of encoding non-stationarity in historical sequence ensemble-based forecasting.

Finally, an assessment of cumulative water supply forecasting skill across 6- and 12-month horizons (based on a visual comparison between 90% prediction intervals and observations in Figures 3 and 4) indicates that our new copula-based forecasts (both with and without matching to initial conditions) capture the strong seasonality of cumulative 6-month water supplies and much of the range of cumulative 12-month water supplies (results across 1- and 3-month horizons shown in Supplementary Information). Forecast stochasticity is apparent in both implementations. Even the forecasts without matching to initial conditions vary somewhat from year to year, an effect that would be diminished with increased sample size; its persistence here (with $n' = 1000$) is indicative of the high variablity in 12-month water supplies. Quantitatively, we find that both implementations of the copula framework (i.e. with and without antecedent water supply matching) are slightly underdispersed in many forecast configurations (Table 1). A closer inspection of our results indicates that many of the historical water supply sequences exceeding the upper and lower bounds of our copula forecast between 2007 and 2021 are associated with extreme hydrological or meteorological events. These include the historic 2011 flooding in areas within and surrounding the Great Lakes basin (Goodwell et al., 2014) followed by a major drought in 2012 (Mallya et al., 2013), along with subsequent years of extreme water abundance across all of the Great Lakes, particularly in 2014 and 2015 on Lakes Superior and Michigan-Huron, 2017 on Lake Ontario, and 2019 on Lake Michigan-Huron (Gronewold et al., 2016; Steinschneider et al., 2019).

Given that our copula is fit to historical conditions before 2007, it is perhaps not surprising that it does not fully capture these extreme events (Gronewold et al., 2021). In future applications (including operational implementation for large lakes and reservoirs), our new system would ideally be re-calibrated to include the most recent hydroclimate data and represent new plausible ranges accordingly. Furthermore, periods where the copula model does not capture extreme events could be improved through manipulation of the model parameters, an approach demonstrated in a preliminary application for multidecadal simulations in VanDeWeghe et al. (2022). Of course, improvements from this approach are dependent on accurate seasonal boundary conditions from regional climate models, one of the primary objectives of the Bipartisan Infrastructure Law (BIL) seasonal-to-annual (S2A) forecasting initiative (Charnley et al., 2023).

The fraction of observations within the 90% prediction interval, shown in Table 1, for the 'matched' ensembles (0.72-0.92) is very similar to that of the random ensembles (0.73-0.90). This result is not entirely surprising because we implement (as a demonstration) only one of many possible 'matching' algorithms for the encoded water balance variables, which themselves may not be the best representation of initial conditions (see further discussion in Section 4). However, both the 'matched' and 'unmatched' ensembles perform either similar to or better than current operational 6-month forecast ensembles (Fry et al., 2020), with model performance generally increasing from Lake Superior to Lake Ontario. Most importantly, the copula modeling approach predicts water supply on an annual time-frame without significant degradation to model performance compared to seasonal forecasts, a milestone not yet achieved in water supply forecasting on the Great Lakes.

4 Discussion and Conclusions

We have presented one possible setting (Great Lakes, seasonal to annual) and implementation framework (R-vine, 'matched' sampling) for water supply forecasting using copulas. In this implementation, the model captures relationships between water balance components and water supply across seasons and across lakes, which is crucial to accurate probabilistic forecasts. Furthermore, we have shown that this approach can be extended to 12-month forecasts which perform comparably to operational forecasts developed for significantly shorter (3- and 6-month) time scales. We acknowledge that the full utility of our copula lake (and reservoir) water supply forecasting system, as with any related lake water supply system, is typically only recognized when coupled with a lake outflow or lake-to-lake routing scheme. For the Great Lakes, there is a long history of developing and testing these routing models, and we leave propagation of our water supply forecasts into Great Lakes water levels for future research.

We have also introduced and demonstrated practical application of an antecedent condition matching scheme that builds off of previous work in water resources management and hydrological forecasting (Wood & Lettenmaier, 2008; Yossef et al., 2013). In this study, our scheme sometimes shifted the uncertainty bounds of S2A forecasts, but did not not ultimately lead to a significant improvement in skill. We believe that future research building on the copula framework should focus not only on refining these initial condition matching algorithms, but also on encoding plausible climatological signatures in the model parameters (such as the mean and variance of future monthly precipitation, as one example). It is also entirely possible to explicitly represent related variables (such as teleconnection indices, soil moisture, temperature, ice cover, etc.) in the copula framework for more refined conditioning to both initial and anticipated future states.

True conditional sampling, however, is one element of copula-based hydro-climate modeling in high dimensions that is not yet fully resolved. While it is possible to constrain the fitting of a copula to a structure that can be used for conditionally sampling a subset of specified variables, the resulting trade-off between the goodness-of-fit of the distribution and the benefits gained from conditional sampling is unclear. Future research efforts are needed to assess this trade-off and the utility of alternative 'matching' processes that can act as proxies for conditional sampling.

While we demonstrate the copula approach with seasonal to annual forecasting, achieving long-term forecasts for assessing trends in water supply dynamics is also entirely feasible. Year-long hydro-climate realizations could be assembled into multi-year sequences which are entirely internally consistent by either conditioning or 'matching' samples based on an overlapping period analogous to the 3-month period of initial conditions presented here. We view implementing copula-based water balance models in this way, including dynamic climate representation (VanDeWeghe et al., 2022), as a very promising alter-

native to using computationally expensive, climate-model-coupled process-based approaches to generate long-term water supply ensembles.

The primary point of this paper is to introduce copulas as a tool for overcoming common challenges in seasonal-to-annual scale water supply forecasting, specifically for large lakes and reservoirs. Copulas provide a framework which is more parsimonious than post-processed resampling methods, and more physically based than most machine learning approaches, while allowing great flexibility for forecasting across multiple timescales and incorporating known or anticipated conditions. Furthermore, copulas (likely vine-copulas) create a basis for developing probabilistic models which 1) inherently capture observed dependencies (flexible to asymmetries, tail-weights, etc.) across dimensions, and 2) allow for unique formulations of the marginal distributions for each individual variable. In the case of water balance forecasting, we have found that these conditions are necessary to appropriately resolve the joint likelihood of multiple hydro-climate realizations across time and space, and the failure to do so will lead to forecasts which are not consistent with the underlying processes governing the generation of water supply.

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6 Conflict of Interest

The authors declare no conflict of interest.

7 Data Availability

The authors will deposit data and model code in a publicly available repository prior to acceptance. For the purposes of review, the data and model code are included in the Supplementary Information file.

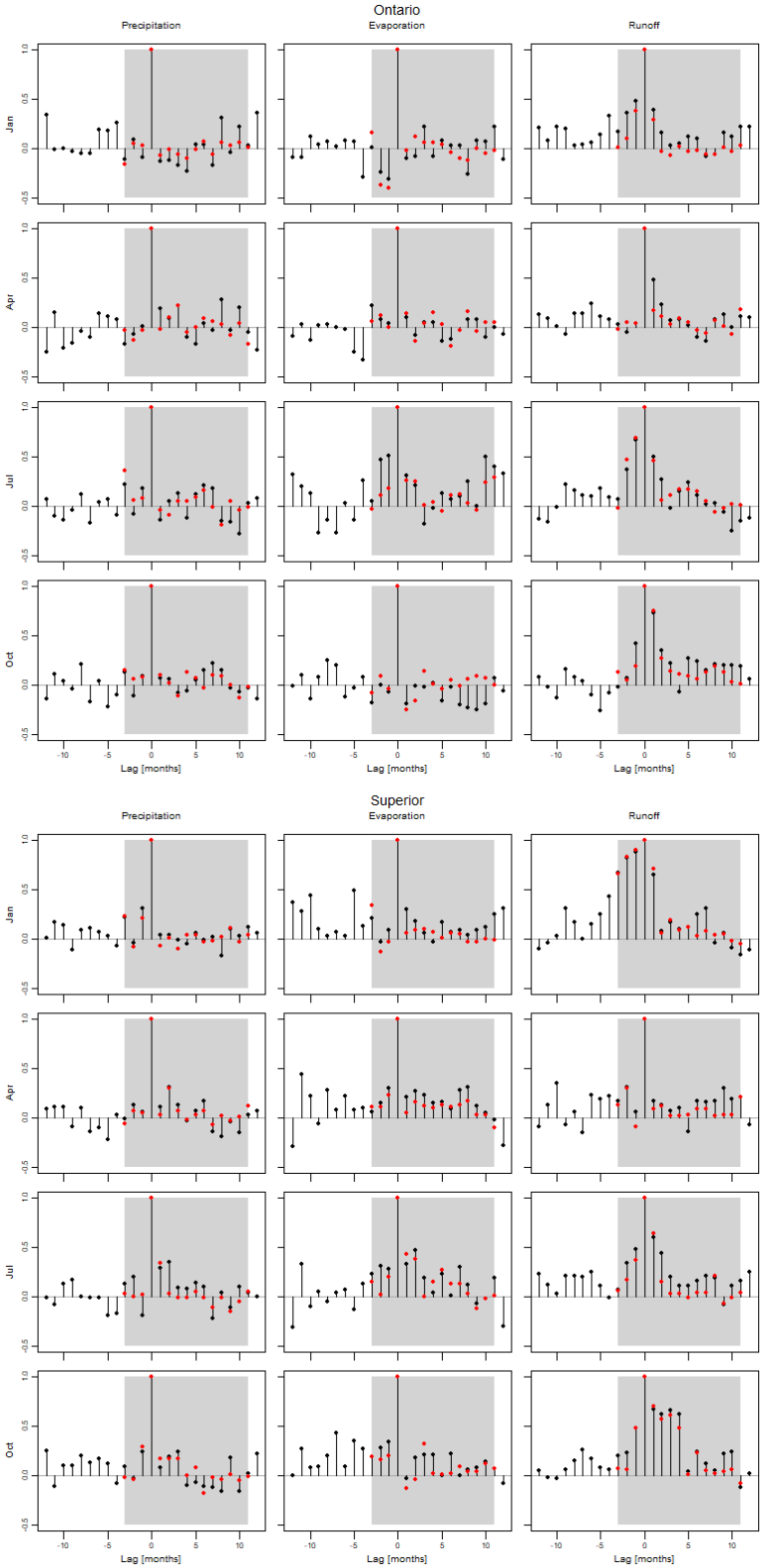

Figure 1. Serial correlation in monthly Great Lakes water balance components from historical data (black) and copula simulations (red). For clarity, only select months, representative of each season, and Lakes (Superior and Ontario) are shown; results for Lakes Michigan-Huron and Erie are included in the supporting information. The grey region highlights a 15-month calibration window which includes a 3-month 'pre-forecast' period for matching initial basin conditions followed by a 12-month forecast horizon.

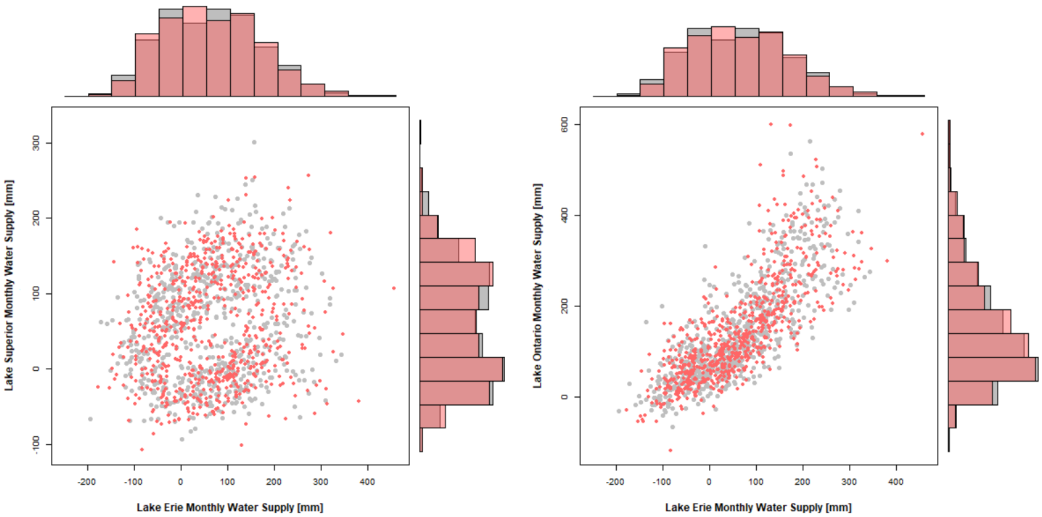

Figure 2. Spatial correlation in monthly Great Lakes water supplies from historical data (grey) and copula simulations (red). For clarity, representative results are shown only for correlation between Lake Superior and Lake Erie (left), and Lake Ontario and Lake Erie (right). Spatial correlation plots across all Great Lakes for 1-, 3-, 6-, and 12-month time frames are included in the Supporting Information. Histograms of Lake Superior, Ontario, and Erie monthly water supplies are presented in panel margins.

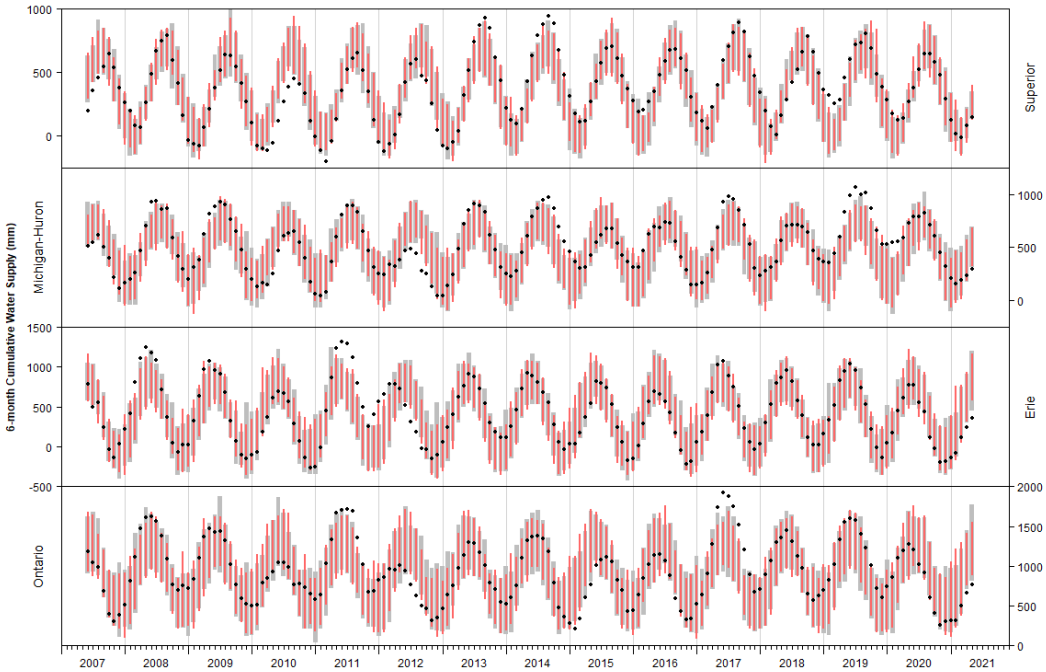

Figure 3. Comparison between observed (black dots) and simulated (grey and red vertical bands) 6-month cumulative water supplies across all of the Great Lakes from Jun 2007 through May 2021 (months represent the end of each 6-month cumulative supply period). Simulated cumulative water supplies are presented as 90% credible intervals for ensembles encoded with (red) and without (grey) matching to initial conditions.

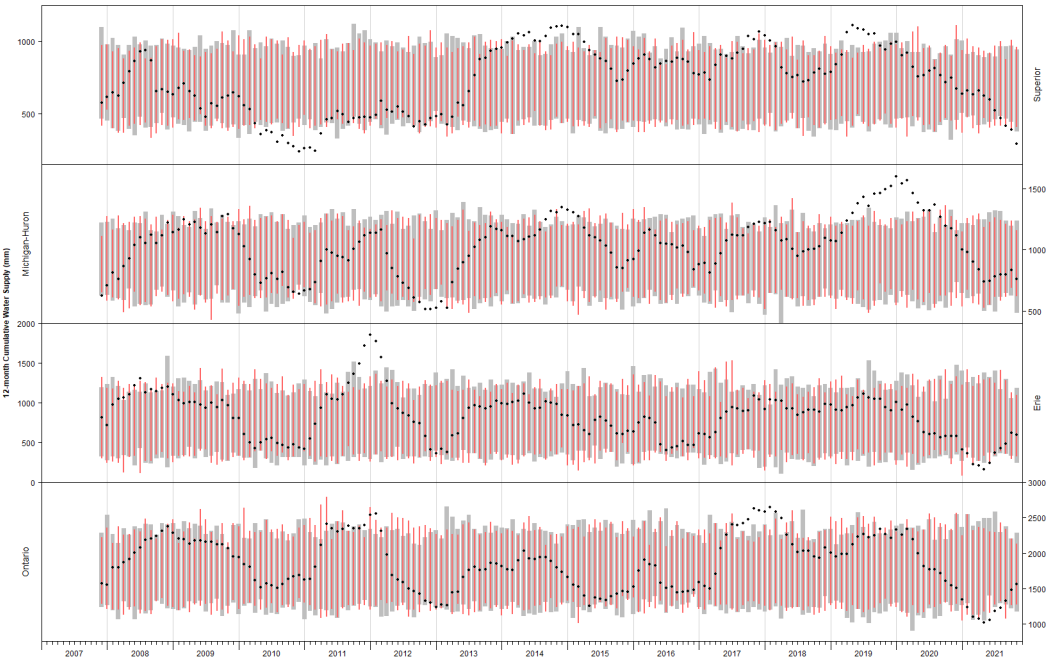

Figure 4. Comparison between observed (black dots) and simulated (grey and red vertical bands) 12-month cumulative water supplies across all of the Great Lakes from December 2007 through November 2021 (months represent the end of each 12-month cumulative supply period). Simulated cumulative water supplies are presented as 90% credible intervals for ensembles encoded with (red) and without (grey) matching to initial conditions.

Table 1. Proportion of observed cumulative water supply values within corresponding 90% prediction intervals from the copula framework.

Lake	1-month	3-month	6-month	12-month
Unmatched to antecedent components				
Superior	0.83	0.77	0.77	0.73
Michigan-Huron	0.82	0.77	0.76	0.78
Erie	0.86	0.86	0.85	0.89
Ontario	0.90	0.86	0.86	0.86
Matched to antecedent components				
Superior	0.82	0.77	0.73	0.72
Michigan-Huron	0.86	0.77	0.77	0.78
Erie	0.87	0.83	0.84	0.92
Ontario	0.87	0.85	0.85	0.85

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